High-Level Design – Reports

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** High-Level Design (sub-module 10 – RPT) · **Version:** 1.0 · **Status:** Draft
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
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> **Based on:** BRD – Reports v1.0, FRD – Reports v1.0 (FR-RPT-01 – FR-RPT-42), TDD v2.0, TaskFlow as built
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version of the Reports HLD (Task Reports screen, summary totals and CSV export), written from the Reports BRD / FRD and the code as built

Related documents

Document	Location
HLD – Overall (stack, deployment, cross-cutting design)	`../../00-Overall/HLD-TaskFlow-Overall.md`
BRD – Reports	`../../../BRD/05-Reports/05.1-Reports/BRD-Reports.md`
FRD – Reports	`../../../FRD/05-Reports/05.1-Reports/FRD-Reports.md`
LLD – Reports (classes, endpoints, queries, tests)	`../../../LLD/05-Reports/05.1-Reports/LLD-Reports.md`
HLD – Dashboard (charts and scoped analytics)	`../../01-Dashboard/01.1-Dashboard/HLD-Dashboard.md`
HLD – Task Management, Project Management (filter-list sources)	`../../02-TaskFlow-Monitoring/02.1-Task-Management/HLD-Task-Management.md`, `../../03-TaskFlow-Management/03.2-Project-Management/HLD-Project-Management.md`
TDD	`TF - Documents/TF - TDD/TDD-TaskFlow.md` (v2.0) – §2.3, §6.3, §6.6, §7.5, §8.7

1. Introduction

1.1 Purpose

This HLD describes how the Reports sub-module is built: its place in TaskFlow, its frontend and backend components, the data it reads, how it uses Task Management and Project Management, its

key flows (summary and CSV export), its security model and the design decisions behind it. Classes, methods, queries, columns and the full endpoint detail are in the LLD.

1.2 Scope

In scope: the **Reports** menu item and the Task Reports screen (`/reports`), the summary totals (Total, Completed, Pending, Overdue) and completion summary, the filters (deadline range, employee, project, status, priority) and the CSV export `taskflow-report.csv`. Out of scope: charts and scoped analytics (Dashboard HLD), workload views (Employee Workload HLD), and the shared stack, deployment and cross-cutting design (overall HLD).

1.3 Audience

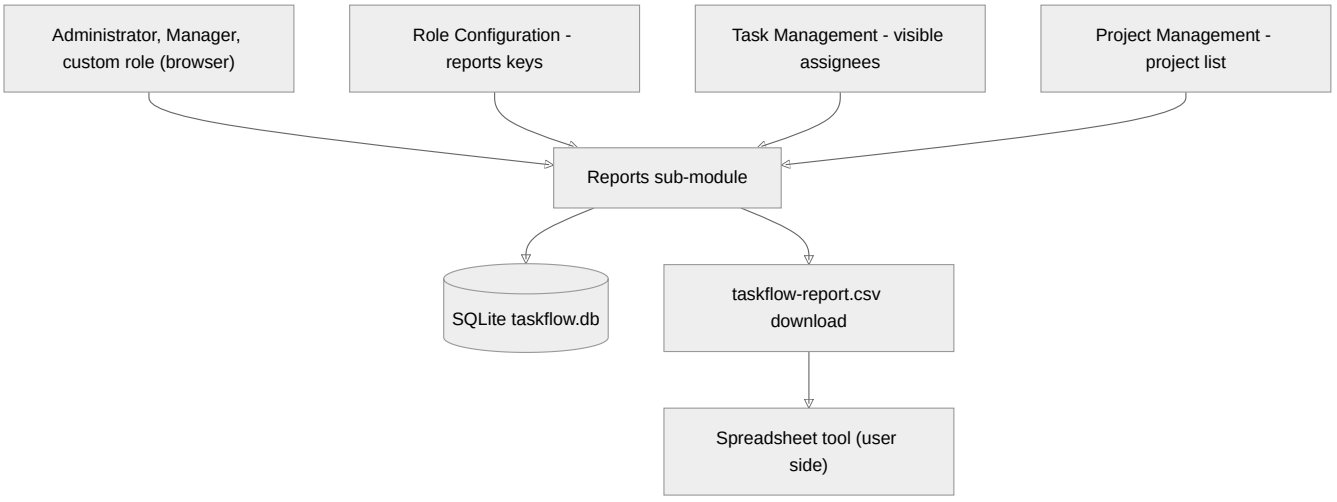
Developer and reviewer (Jojo George), technical reviewers, and anyone maintaining reporting or exports.

1.4 Definitions & Abbreviations

Term	Meaning
SPA	Single Page Application (Angular frontend)
Summary	The four totals and the completion percentage for the current filter
Export	The filtered task rows written as a CSV file
CSV	Comma-separated values (RFC 4180 style, CRLF line ends, header row)
Formula protection	Writing a leading `'` before a cell starting with `=`, `+`, `-` or `@` so spreadsheets do not run it
Allocation	Link between an employee and a project (`UserProjectAllocation`)
Organisation scope	Every holder of `reports.view` sees all non-deleted tasks

2. System Context

2.1 Context Diagram



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2.2 Actors & External Systems

Actor / system	Interaction	Protocol
Administrator, Manager	Open Task Reports, filter, export	HTTPS (SPA → REST JSON; CSV download)
Custom role with `reports.view`	Same as above; export only with `reports.export`	HTTPS
Employee	No access by default	–
Task Management (internal)	Supplies the Employee filter list (assignees the user can see)	REST JSON
Project Management (internal)	Supplies the Project filter list (needs `projects.view`)	REST JSON
Role Configuration (internal)	Grants `reports.view` and `reports.export`	Database (role permissions)
Spreadsheet tool (user side)	Opens the downloaded CSV; not called by TaskFlow	File

No external (third-party) service is called by this sub-module.

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3. Architecture Goals, Constraints & Principles

3.1 Quality Goals

Goal	How the architecture supports it
Security	View and export are separate permission keys checked by the server; export requires both
Safe output	The CSV writer applies formula protection and proper quoting to every cell
Correct totals	Overdue is derived at query time from status and deadline using the application clock (system decision D5)
History preservation	Soft-deleted tasks are left out; tasks of removed employees keep the person's name (system decision D6)
Performance	Totals come from one aggregate query; the export is capped at 50,000 rows
Maintainability	Controller → Service → Repository with a validator and a separate CSV writer

3.2 Constraints

- Single SQLite database shared with all modules; no reporting replica.
- Export is built in memory and capped at 50,000 rows (code constant); only CSV is offered.
- Fixed permission catalogue: `reports.view`, `reports.export` (export requires view).
- No department filter and no per-user data scope in this release (BRD §3.2).

3.3 Architecture Principles

- One filter model for summary and export, so the file always holds the tasks behind the totals.
- Server-side validation of every filter value; the UI date pickers only help.

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- Read-only module: no write, no notification, no mail, no activity entry.
- Output encoding at the edge: the CSV writer is the only place that formats cells.

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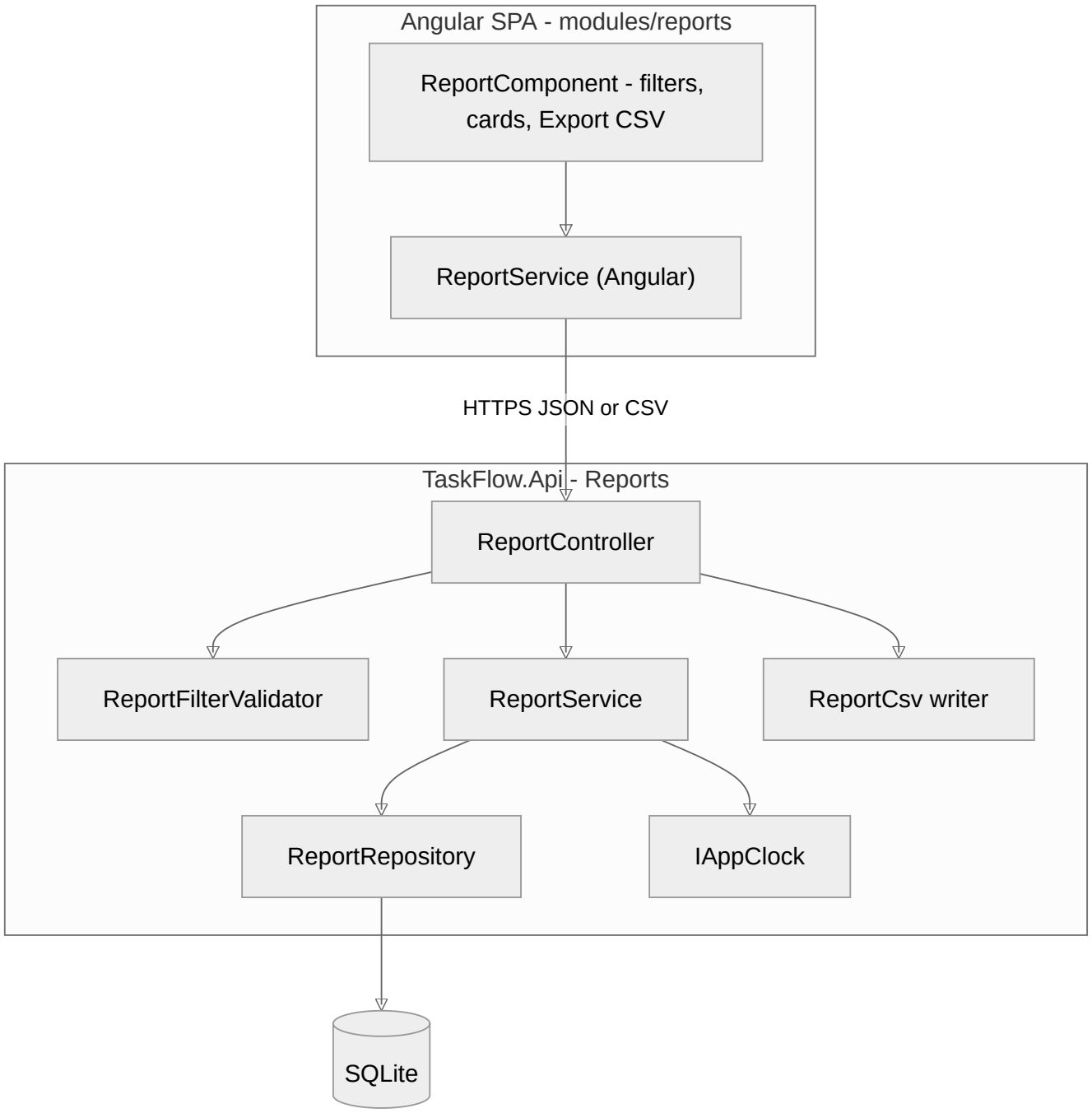
4. Architecture Overview

4.1 Architecture Style

Reports follows the system-wide style – Angular SPA calling a layered ASP.NET Core REST API over SQLite – described in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`), which also covers the technology stack, deployment, the request pipeline (JWT, session validation, MFA gate, permission policies, validation filter) and logging. This section only shows the Reports slice.

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4.2 Logical Architecture



4.3 Layers & Responsibilities

Layer	Reports responsibility	May call
Presentation	Route guard, menu item, filter toolbar, count cards, completion bar, export button, blob download, error panel and toast	Reports API, Task and Project APIs (filter lists)
API / Controller	Bind the filter, enforce `reports.view` / `reports.export`, return JSON or a CSV file	Validator, service, CSV writer
Validation	Reject a reversed deadline range and undefined status / priority values	–

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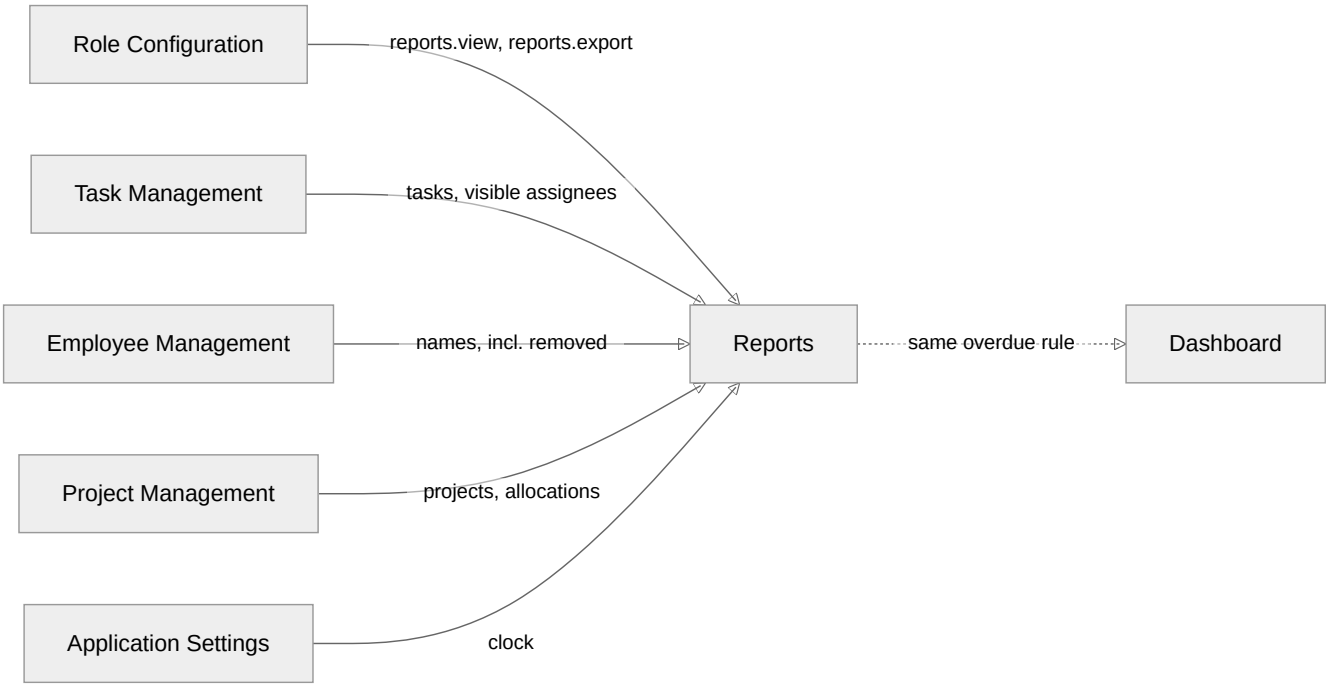
Layer	Reports responsibility	May call
Service	Map the filter to criteria (UTC bounds), compute totals at the current time, build export rows with the assignee's projects	Repository, `IAppClock`
Repository	Read-only queries: aggregate totals, export rows, project names per user	Database
CSV writer	Header, escaping, formula protection, date format	–

5. Component / Module Design

Component	Responsibility	Key components (high level)	Depends on
Reports route and menu (frontend)	Show **Reports** only with `reports.view`; redirect others to their own dashboard	`REPORTS_ROUTES` with `permissionGuard`, `nav-items`	Auth session (permissions), `dashboardRouteFor`
Task Reports page (frontend)	Filters with instant reload, **Reset**, four cards, completion bar, **Export CSV** (shown with `reports.export`), "Exporting..." state, error panel and toast	`ReportComponent`	`ReportService` (Angular), task and project services, shared `local-day`, `save-blob`, toast
Report API client (frontend)	Call summary and export; drop empty filter values; download the CSV as a blob	`ReportService` (Angular)	Reports API
Reports API (backend)	Two read-only GET endpoints: summary and export	`ReportController`	Permission policies, validation filter
Filter validation (backend)	Deadline order; enum values	`ReportFilterValidator` (FluentValidation)	Shared `ValidationFilter`
Report logic (backend)	Totals, export rows, project list per assignee, row cap	`ReportService`	`ReportRepository`, `IAppClock`
CSV writer (backend)	RFC 4180 output with formula protection; file name and media type	`ReportCsv`	–
Report data access (backend)	Aggregate totals, ordered capped rows, allocation names	`ReportRepository`	SQLite via EF Core

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5.1 Module Interaction



Interacting sub-module	Direction	What is exchanged
Role Configuration	In	Grants the two report keys; `reports.export` requires `reports.view`
Task Management	In (read + API)	Task data for totals and rows; `/api/v1/tasks/assignees` for the Employee list (scoped to the caller's task access)
Project Management	In (read + API)	Allocations and names for the Project filter and column; `/api/v1/projects` for the Project list (needs `projects.view`)
Employee Management	In (read)	Assignee names, including removed (soft-deleted) users
Application Settings	In (in-process)	`IAppClock` current UTC time for the overdue rule
Dashboard	Related	Same instant overdue rule; the Dashboard offers charts and scoped analytics, Reports offers totals and the file

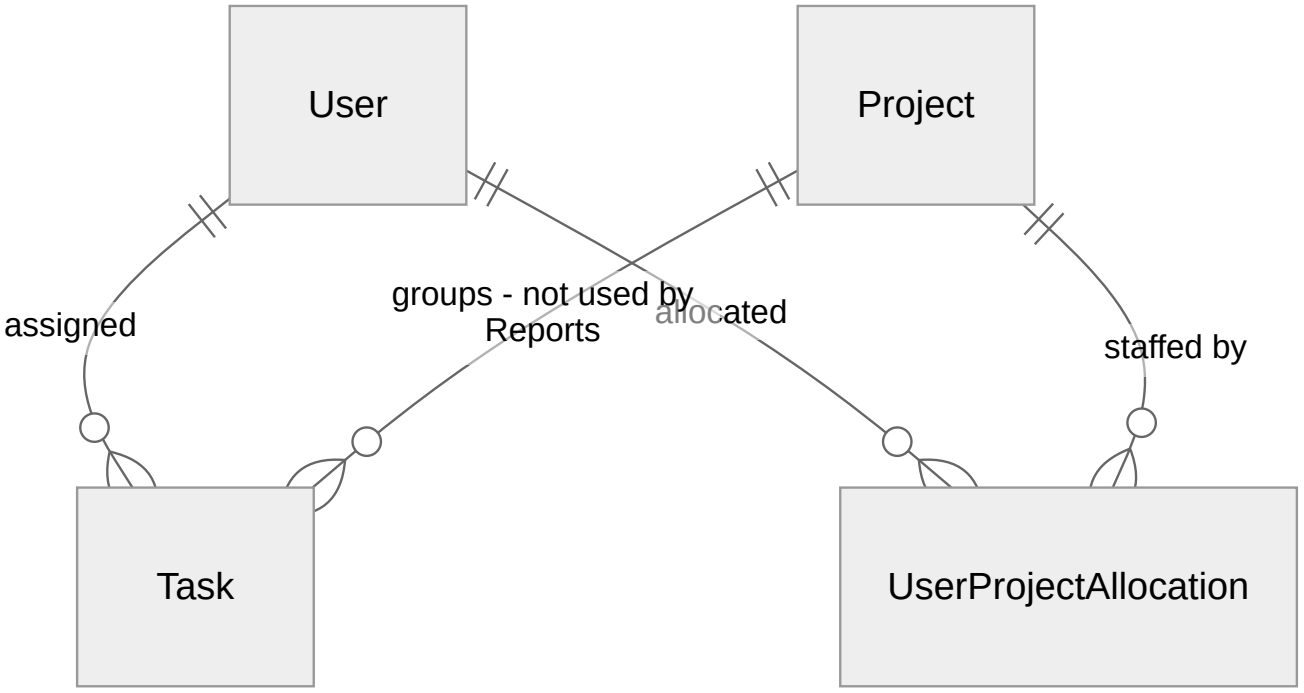
6. Data Architecture (high level)

6.1 Data Stores

Store	Technology	Holds (for this sub-module)	Notes
Main database	SQLite (`taskflow.db`, EF Core)	Tasks, users, project allocations, projects	Read only; Reports owns no table and adds no migration
Browser	Blob download	`taskflow-report.csv` saved by the user	Not kept by TaskFlow

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6.2 Main Data Domains



Domain	Owner sub-module	Use in Reports
Tasks	Task Management	Filters, totals, export rows; deleted tasks excluded
People	Employee Management	Employee filter value and Assignee column; removed users kept
Project allocations	Project Management	Project filter and Project column (assignee's allocations, not the task's own project)
Projects	Project Management	Project names, sorted by name

6.3 Data Flow

1. The page sends the optional filter values; the deadline days are turned into UTC start and end instants by the browser.
2. The validator checks the range order and enum values; the service normalises the bounds to UTC.
3. Summary: the repository returns Total, Completed, Pending and Overdue in one aggregate query, with Overdue measured at the clock's current time.
4. Export: the repository returns up to 50,000 matching rows, earliest deadline first; the service adds each assignee's project names; the CSV writer formats the file.
5. The browser shows the cards and completion bar, or saves the file.

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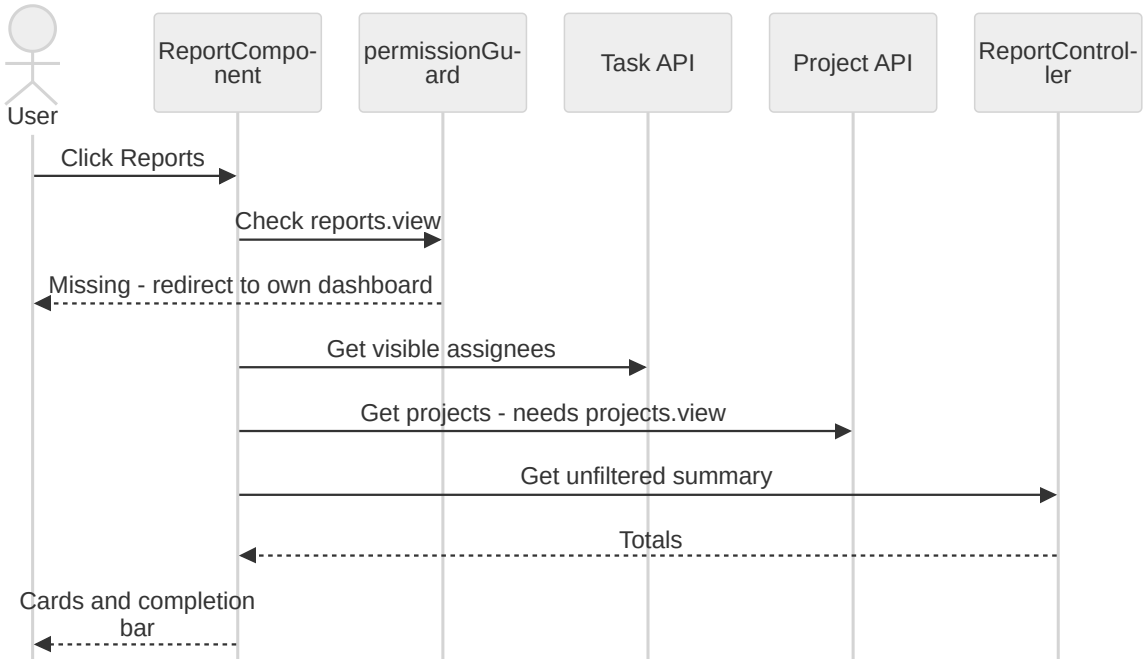
7. Integration Design

Integration	Direction	Purpose	Failure behaviour
Task Management – `/api/v1/tasks/assignees`	Outbound (SPA)	Employee filter options	List stays empty; report still works (no message – R-RPT-04)
Project Management – `/api/v1/projects`	Outbound (SPA)	Project filter options	403 without `projects.view`; list stays empty with no message (R-RPT-04)
Application Settings – `IAppClock`	Inbound (in-process)	Current UTC time for Overdue	–
Browser file download	Outbound (response)	CSV attachment `taskflow-report.csv` (`text/csv`, UTF-8)	Any error → toast "Could not export the report. Please try again."

API surface of the sub-module (details in the LLD): `/api/v1/reports` (summary, JSON) and `/api/v1/reports/export` (CSV file). Both are GET and read-only. No external service is called.

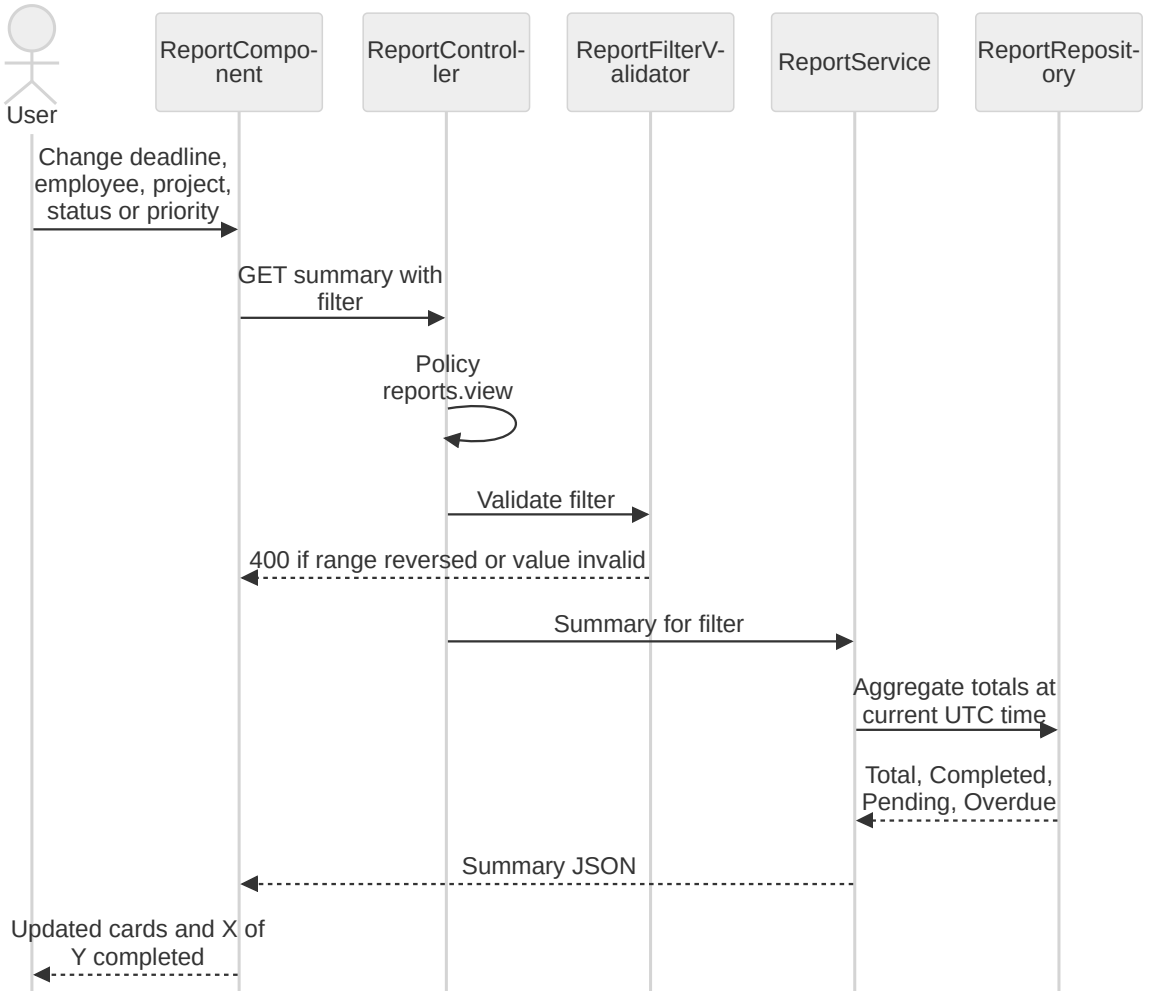
8. Key Flows (high level)

8.1 Open Task Reports (access and filter lists)



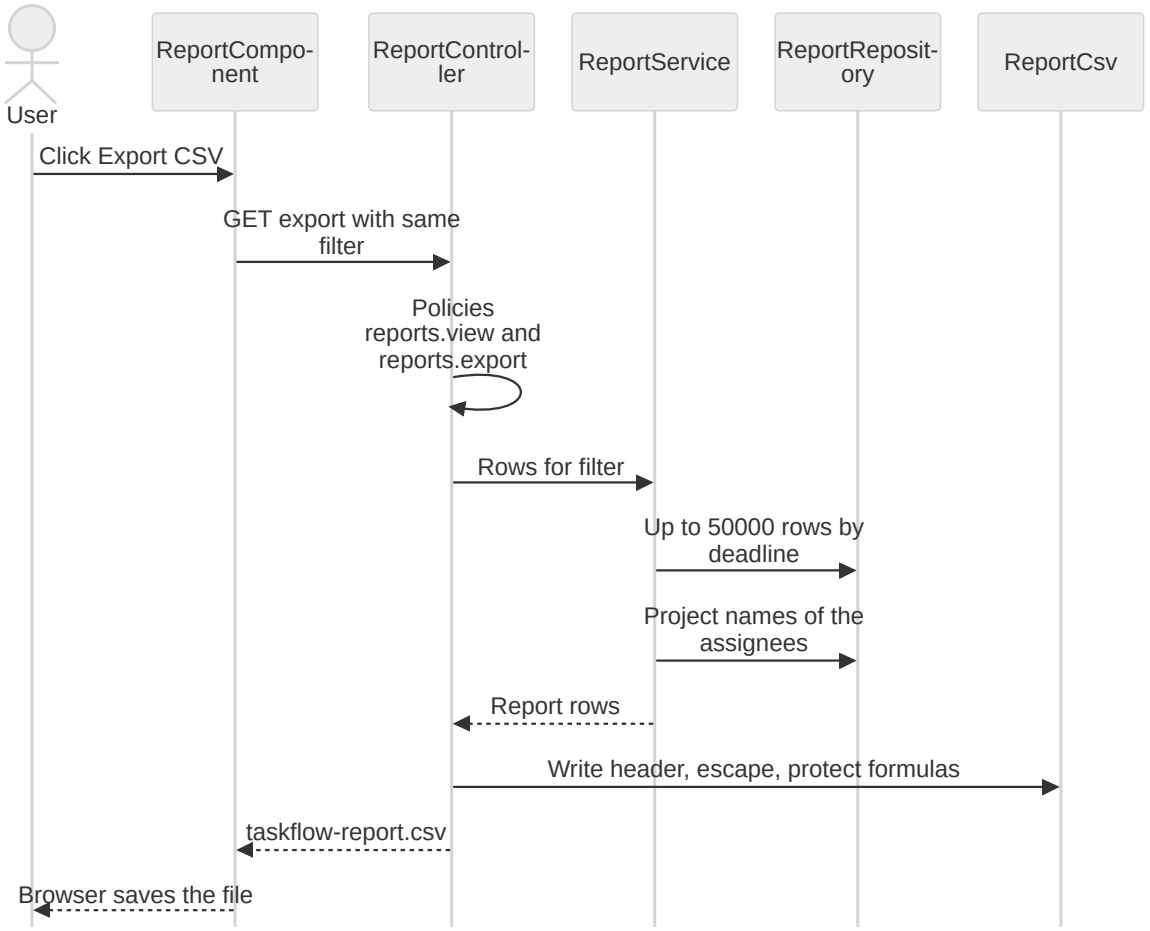
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8.2 Filter the report (main business flow)



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8.3 Export CSV (integration flow)



8.4 Background job

Not applicable for this sub-module. Reports run on request only; there are no scheduled or emailed reports.

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9. Security Architecture

Concern	Approach
Authentication	Shared JWT bearer token + refresh-token cookie with session revalidation and MFA gate (overall HLD)
Authorization	Controller-level `reports.view` covers both endpoints; export additionally needs `reports.export`. Defaults: Administrator and Manager granted, Employee denied, custom role configurable; the catalogue refuses `reports.export` without `reports.view`
Data scope	Organisation-wide for every holder of `reports.view` (no Own / Team narrowing) – see D-RPT-1 and R-RPT-01
Input validation	FluentValidation on the filter (range order, enum values); model binding rejects unparseable values (400)
Output safety	Formula protection and quoting in the CSV writer (NFR-RPT-02)

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Concern	Approach
Frontend	Route guard and menu item on `reports.view`; **Export CSV** button only with `reports.export`. UI hiding is a convenience; the server is the authority
Data protection	Export holds task and person names only; no secrets; nothing stored server-side
Auditing	Refused requests (403) are logged by the shared denied-action audit handler (user, method, path, policy); views and exports create no activity entry

10. Technology Stack

The shared stack (ASP.NET Core / .NET 10, EF Core 10.0.12 with SQLite, Angular 19.2, JWT) is described in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`). Sub-module-specific items:

Area	Technology	Version	Reason
Filter validation	FluentValidation	12.2.0	Same validation pipeline as the rest of the API
CSV output	In-house writer (`ReportCsv`) on .NET	net10.0	Small, dependency-free, with formula protection
Data access	EF Core LINQ aggregate on SQLite	10.0.12	Totals computed in one query
File download	Angular `HttpClient` blob + shared save helper	Angular ^19.2.0	Keeps the bearer token on the request; no public file URL

11. Deployment Architecture

Not applicable for this sub-module. Reports run inside the single `TaskFlow.Api` process and the Angular SPA; they add no process, port, folder or configuration section. Deployment and environments are in the overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`). Configuration read: `ConnectionStrings:Default` (`<secret>`) and the shared `Jwt:*` lifetimes; the 50,000-row cap, file name and media type are code constants.

12. Non-Functional Design

Area	Design decision
Performance	Summary in one aggregate, no-tracking query using the deadline and assignee indexes; export capped at 50,000 rows. > ⚠️ To be confirmed: a measured response-time target (NFR-RPT-06)
Scalability	Single API instance on SQLite; export built in memory, bounded by the row cap (NFR-RPT-04)
Reliability	Stateless, read-only requests; a failed export can simply be repeated
Availability	Same as the API process (overall HLD)

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Area	Design decision
Security	Server-side view / export checks with audit of refusals (NFR-RPT-01); formula protection (NFR-RPT-02)
Compatibility	Header row, comma separator, quoting, CRLF, `yyyy-MM-dd` dates (NFR-RPT-03)
Usability	Instant reload on filter change, **Reset**, plain-language error panel and toast (NFR-RPT-05)
Maintainability	Controller → Service → Repository; CSV writer and validator as separate units with their own tests
Observability	Shared request logging and denied-action audit
Backup & Recovery	Nothing owned; covered by the data-folder backup in the overall HLD

13. Design Decisions

ID	Decision	Alternatives considered	Rationale
D-RPT-1	Organisation-wide data for every holder of `reports.view`	Own / Team scope per caller (original BRD FR6)	Matches the dashboards' organisation view for report users; access is controlled by who holds the key. > ⚠️ To be confirmed: acceptance for Managers (R-RPT-01)
D-RPT-2	Separate `reports.view` and `reports.export` keys; export requires view	One key for both	Administrators can allow on-screen totals without allowing data to leave the system
D-RPT-3	One filter model shared by summary and export	Separate export parameters	The file always contains the tasks behind the totals
D-RPT-4	Totals in one aggregate query; Overdue derived at the current clock time	Stored overdue flag; several count queries	Never stale (system decision D5); one round trip
D-RPT-5	CSV built in memory, capped at 50,000 rows, earliest deadline first	Streamed export; no cap; Excel / PDF	Simple and bounded for a single-server deployment; CSV opens everywhere. > ⚠️ To be confirmed: warn when the cap is reached (R-RPT-02)
D-RPT-6	Formula protection and quoting in a dedicated CSV writer	Plain string join; third-party CSV library	Prevents spreadsheet formula injection; no extra dependency
D-RPT-7	Project filter and column use the assignee's allocations	The task's own project	Reports what each person is staffed on, as built. > ⚠️ To be confirmed: use the task's project (R-RPT-06)
D-RPT-8	Keep removed (soft-deleted) users' tasks and names; exclude deleted tasks	Hide everything linked to removed users	Finished work still counts (system decision D6, BR-RPT-06)
D-RPT-9	Deadline days converted to UTC instants in the browser; server treats unspecified dates as UTC	Convert in the application timezone on the server	Simple, as built. > ⚠️ To be confirmed: move to the application timezone (R-RPT-03)

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14. Risks, Assumptions & Open Issues

Type	Description	Mitigation / owner
Risk	Managers see tasks outside their team (organisation scope) (R-RPT-01)	Grant `reports.view` only to trusted roles. > ⚠️ To be confirmed: per-user scope – Jojo George
Risk	Exports over 50,000 rows are cut without a warning (R-RPT-02)	Narrow the filters. > ⚠️ To be confirmed: warning – Jojo George
Risk	Deadline-day bounds follow the browser's time zone, not the application timezone (R-RPT-03)	> ⚠️ To be confirmed – Jojo George
Risk	Project / Employee list stays empty without `projects.view` or on a load failure, with no message (R-RPT-04)	Grant `projects.view` with `reports.view`. > ⚠️ To be confirmed: message – Jojo George
Risk	Employee list follows Task List access while totals are organisation-wide (R-RPT-05)	Review custom role grants – Jojo George
Risk	Reports count Cancelled tasks in Total and all open tasks as Pending, while the dashboards exclude Cancelled and split Pending / In progress (BR-RPT-10)	> ⚠️ To be confirmed: align – Jojo George
Risk	The server's validation message is not shown; the screen shows the generic load error (the date pickers normally prevent the case)	> ⚠️ To be confirmed – Jojo George
Assumption	Data volume fits on-request aggregation and in-memory export on SQLite	Re-check if task volume grows
Open issue	> ⚠️ To be confirmed: measured response-time target (NFR-RPT-06)	Jojo George

15. Traceability (requirements → components)

FRD item(s)	Module / component	Section
FR-RPT-01, FR-RPT-02	Reports route and menu (`REPORTS_ROUTES` , `permissionGuard` , `nav-items`)	§5, §8.1
FR-RPT-03	Reports API (`ReportController` , `reports.view` policy, denied-action audit)	§5, §9
FR-RPT-04, FR-RPT-05, FR-RPT-06	Task Reports page (`ReportComponent`) + filter-list integrations	§5, §7, §8.1
FR-RPT-07	Task Reports page (count cards)	§5
FR-RPT-08, FR-RPT-09, FR-RPT-10, FR-RPT-11, FR-RPT-12, FR-RPT-14	Report logic (`ReportService`) + `ReportRepository` aggregate + `IAppClock`	§5, §8.2, D-RPT-4
FR-RPT-13	Task Reports page (completion bar)	§5
FR-RPT-15, FR-RPT-16	Page date pickers (local day to UTC) + `ReportService` bounds + repository filter	§6.3, D-RPT-9
FR-RPT-17	Task Reports page (date picker limits)	§5

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FRD item(s)	Module / component	Section
FR-RPT-18	Filter validation (`ReportFilterValidator`)	§8.2, §9
FR-RPT-19, FR-RPT-23, FR-RPT-24, FR-RPT-25	`ReportRepository` filters; validator for enum values	§5, §9
FR-RPT-20	Task Management integration (`/api/v1/tasks/assignees`)	§7
FR-RPT-21	`ReportRepository` allocation filter	§6.2, D-RPT-7
FR-RPT-22	Project Management integration (`/api/v1/projects`, `projects.view`)	§7
FR-RPT-26, FR-RPT-27	Task Reports page (instant reload, **Reset**)	§5
FR-RPT-28, FR-RPT-29	`ReportRepository` (deleted tasks out, removed users kept)	§6.2, D-RPT-8
FR-RPT-30	`ReportService` (organisation scope)	§9, D-RPT-1
FR-RPT-31	Task Reports page (**Export CSV** with `reports.export`)	§5, §9
FR-RPT-32	Reports API (`reports.view` + `reports.export` policies, audit)	§8.3, §9, D-RPT-2
FR-RPT-33, FR-RPT-34	`ReportService` + `ReportRepository` (shared filter, order, 50,000 cap)	§8.3, D-RPT-3, D-RPT-5
FR-RPT-35, FR-RPT-36, FR-RPT-38, FR-RPT-39	CSV writer (`ReportCsv`)	§8.3, D-RPT-6
FR-RPT-37	`ReportService` project names per assignee	§8.3, D-RPT-7
FR-RPT-40, FR-RPT-41	Task Reports page + Angular `ReportService` (blob download, toast)	§5, §7
FR-RPT-42	Whole sub-module (read-only, no side effects)	§3.3

All 42 FRs (FR-RPT-01 – FR-RPT-42) are traced to at least one component.

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